

1 General Renewal Process

1.1 Asymptotic Distribution

Consider the following signal

$$u(t) = \chi_k, \text{ with } t \in [T_k, T_{k+1}], \text{ and } \chi_{k+1} = -\chi_k \quad (1)$$

The times T_i are considered independent and identically distributed random variables with expected mean value $E[T_i] = \mu$, variance $V[T_i] = \sigma$, probability density $q(t)$ and cumulative probability density $Q(t)$, while $N(t)$ is the number of events inside the interval $[0, t)$. We also consider the waiting time to the n -th event.

$$G_n = T_1 + T_2 + \dots + T_n \quad (2)$$

such that

$$\text{Prob}\{G_n > t\} = \text{Prob}\{N(t) < n\} \quad (3)$$

It can be shown that the distribution of $N(t)$ as $t \rightarrow \infty$ is asymptotically normal with expected mean value $E[N(t)] = t/\mu$ and variance $V[N(t)] = \sigma^2 t/\mu^3$.

1.2 Relation between the autocorrelation function and the power spectrum

In applied mathematics, the Wiener-Khinchin theorem, states that the autocorrelation function of a wide-sense-stationary random process has a spectral decomposition given by the power spectrum of that process

$$\text{PSD}(\omega) = \int_{-\infty}^{\infty} C(t) \exp(-i\omega t) dt, \text{ with } \omega = 2\pi f \quad (4)$$

where the autocorrelation function reads as

$$C(t) \equiv \frac{1}{T_f} \int_0^{T_f} u(t')u(t+t')dt' \quad (5)$$

The value $u(t')u(t+t')$ may be written as $\chi_k\chi_l$, where $t' \in [T_l, T_{l+1})$ and $t+t' \in [T_k, T_{k+1})$. The idea is to approximate the autocorrelation function using the probability distribution $q(\tau)$ by considering only the time intervals where $\chi_k\chi_l > 0$,

$$C(t) \approx \frac{1}{\mu} \int_t^{T_f} (\tau - t)q(\tau)d\tau \quad (6)$$

Let us suppose that $q(\tau) = K\tau^{-\beta}$ where K is some proportionality constant. Integration of equation (6) leaves

$$C(t) \approx \frac{K}{\mu} \frac{(2-\beta)(1-\beta)}{(2-\beta)(1-\beta)} t^{2-\beta}, \text{ with } \beta > 2 \quad (7)$$

$$\begin{aligned} \text{PSD}(f) &\sim \int_{-\infty}^{\infty} t^{2-\beta} \exp(-i\omega t) dt \\ &\sim f^{\beta-3} \int_{-\infty}^{\infty} u^{2-\beta} \exp(-iu) du \end{aligned}$$

Suggesting that the power spectrum decays as $\alpha = \beta - 3$ for $2 < \beta$. A similar expression may be recovered for $1 < \beta < 2$.

Question: a) Does $q(\tau) \sim \tau^{-\beta}$ fits to any of our observations? b) How about $\text{PSD}(f) \sim f^{-\alpha}$? If so, for which frequency range?

1.3 A model for the probability distribution $q(t)$

Let us consider the survival function as the probability that an event will survive beyond a time t .

$$s(t) = \text{Prob}\{T_i > t\} = 1 - Q(t) \quad (8)$$

Also consider the hazard function as the instantaneous failure rate

$$h(t) = \lim_{dt \rightarrow 0} \frac{\text{Prob}\{t < T_i < t + dt\}}{dt} \frac{1}{s(t)} = \frac{q(t)}{s(t)} \quad (9)$$

also expressed as a cumulative hazard function

$$\Lambda(t) = \int_0^t h(t)dt = -\log(s(t)) \quad (10)$$

The example given by [Herault] consists in two attractors in the phase-space: one attractor A, corresponding to the value $u = +1$, and another attractor B, corresponding to the value $u = -1$. The system moves from one attractor to the other with a given probability $\Gamma(t)$, which corresponds precisely to the hazard function $h(t)$, while the probability function $P(t) = \Gamma(t)f(t)$ corresponds precisely to the expression above $q(t) = s(t)h(t)$.

Let us assume a constant hazard rate $h(t) = \lambda$, meaning the probability of observing a future event does not event on the time passed from the previous event, i.e. the system has no memory of previous events. In such case,

$$h(t) = \lambda \quad (11)$$

$$\Lambda(t) = \int_0^t h(t)dt = \lambda t \quad (12)$$

$$s(t) = \exp(-\Lambda(t)) = \exp(-\lambda t) \quad (13)$$

$$q(t) = s(t)h(t) = \lambda \exp(-\lambda t) \quad (14)$$

one obtains the Exponential distribution.

Now, let us assume a hazard rate $h(t)$ that is uniformly distributed between two values λ_1 and λ_2 ,

$$h(t) = \begin{cases} \frac{1}{\lambda_2 - \lambda_1}, & \text{for } \lambda_1 < t < \lambda_2 \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

$$\Lambda(t) = \int_0^t h(t)dt = \frac{(t - \lambda_1) - (t - \lambda_2)\theta(t - \lambda_2)}{\lambda_2 - \lambda_1} \theta(t - \lambda_1) \quad (16)$$

$$s(t) = \exp(-\Lambda(t)) = \exp\left(-\frac{(t - \lambda_1) - (t - \lambda_2)\theta(t - \lambda_2)}{\lambda_2 - \lambda_1} \theta(t - \lambda_1)\right) \quad (17)$$

$$q(t) = s(t)h(t) = \begin{cases} \frac{1}{\lambda_2 - \lambda_1} \exp\left(-\frac{t - \lambda_1}{\lambda_2 - \lambda_1}\right), & \text{for } \lambda_1 < t < \lambda_2 \\ 0, & \text{otherwise} \end{cases} \quad (18)$$

Now, [Herault] considers a system with some form of memory given by

$$h(t) = \frac{\nu}{\theta + t} \quad (19)$$

$$\Lambda(t) = \int_0^t h(t)dt = -\log \left(\frac{\theta}{\theta + t} \right)^\nu \quad (20)$$

$$s(t) = \exp(-\Lambda(t)) = \left(\frac{\theta}{\theta + t} \right)^\nu \quad (21)$$

$$q(t) = s(t)h(t) = \frac{\nu\theta^\nu}{(\theta + t)^{\nu+1}} \quad (22)$$

In his thesis, [Herault] considers the case where moving from A to B, and moving from B to A, have different probabilities

$$h_{A \rightarrow B}(t) = \frac{\nu_1}{\theta + t}, \text{ and } h_{B \rightarrow A}(t) = \frac{\nu_2}{\theta + t} \quad (23)$$

and uses Monte Carlo simulations to study the case with $\nu_1 = \nu_2$, $\nu_1 < \nu_2$, and $\nu_1 \ll \nu_2$. Two different behaviours are observed: the proposed $\alpha + \beta = 3$ for symmetric configurations, and $\alpha - \beta = 1$ for the strongly asymmetric configurations, the latter being more relevant for burst intermittent dynamics.